

Ethanol Blending in Gasoline – Taiwan

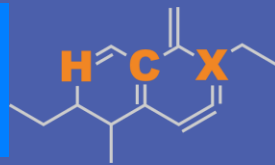
October 2025



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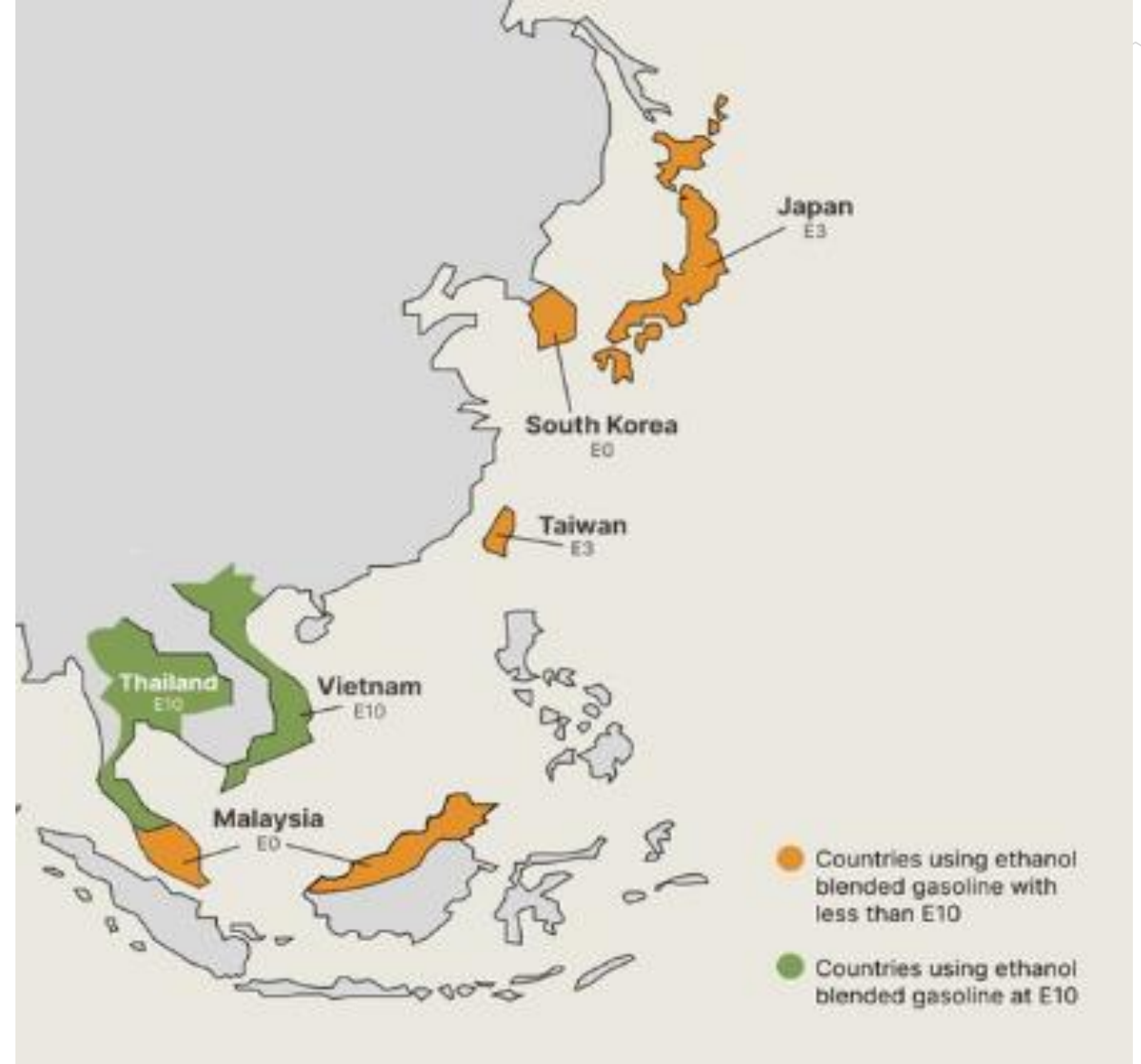


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Ethanol Blending in Asia

There are important fuel quality and environmental impact of vehicle emission challenges in these regions.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Asia to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Seven countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

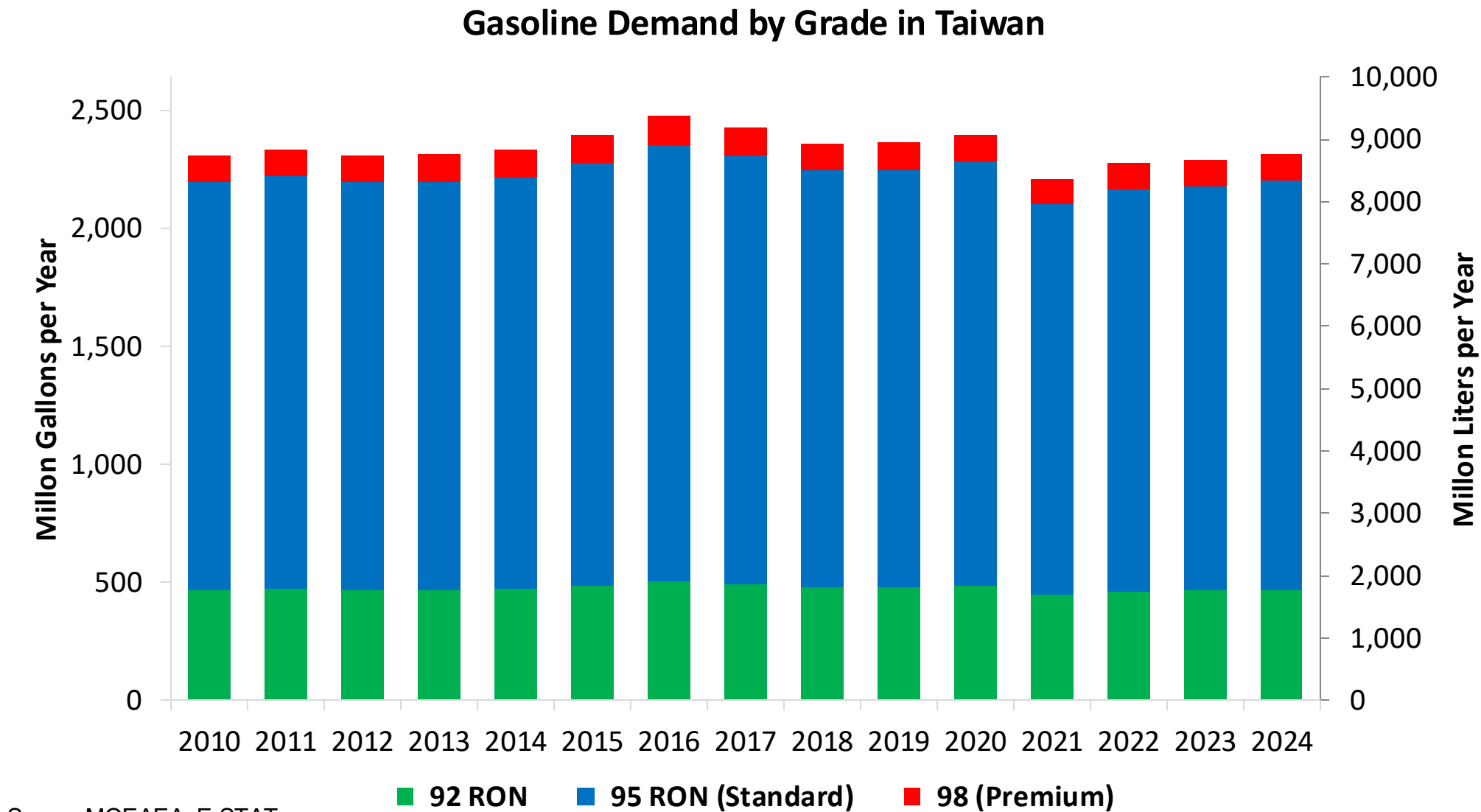
Ethanol Gasoline Blending in Taiwan



In 2024, gasoline consumption in Taiwan was 8,761 million liters (2,314 million gallons) and growth rate was -0.4%. Gasoline production and net imports were 12,450 and -3,689 million liters (3,289 and -975 million gallons) correspondingly. Taiwan offers three main grades: RON 92, RON 95 and RON 98) with an average octane of 94.5 RON.

Taiwan has policies to promote bioethanol in transportation to reduce imported fuel dependency and lower carbon emissions, using it as an oxygenate additive. While there was a limited trial of E3 gasohol in the mid-2000s, the lack of domestic production and insufficient incentives led to its discontinuation. Only a few gas stations offer it. Ethanol is supplied by imports and corresponds to an average content of 0.2%. Future bioethanol use depends on developing profitable cellulosic ethanol production from non-food waste like rice straw and food waste, rather than imported feedstocks.

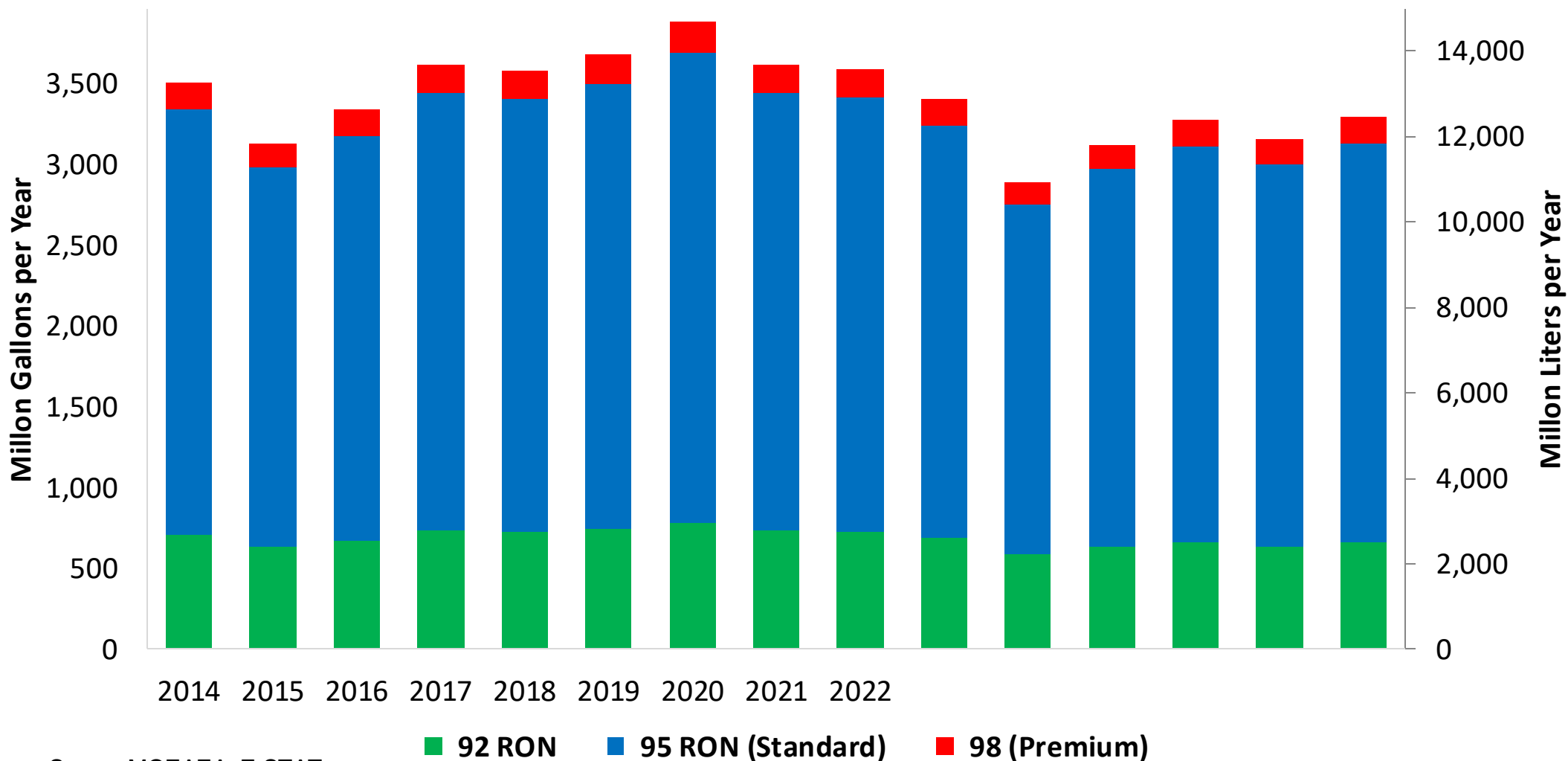
Gasoline Demand in Taiwan



Source: MOEAEA, E-STAT

Gasoline Production in Taiwan

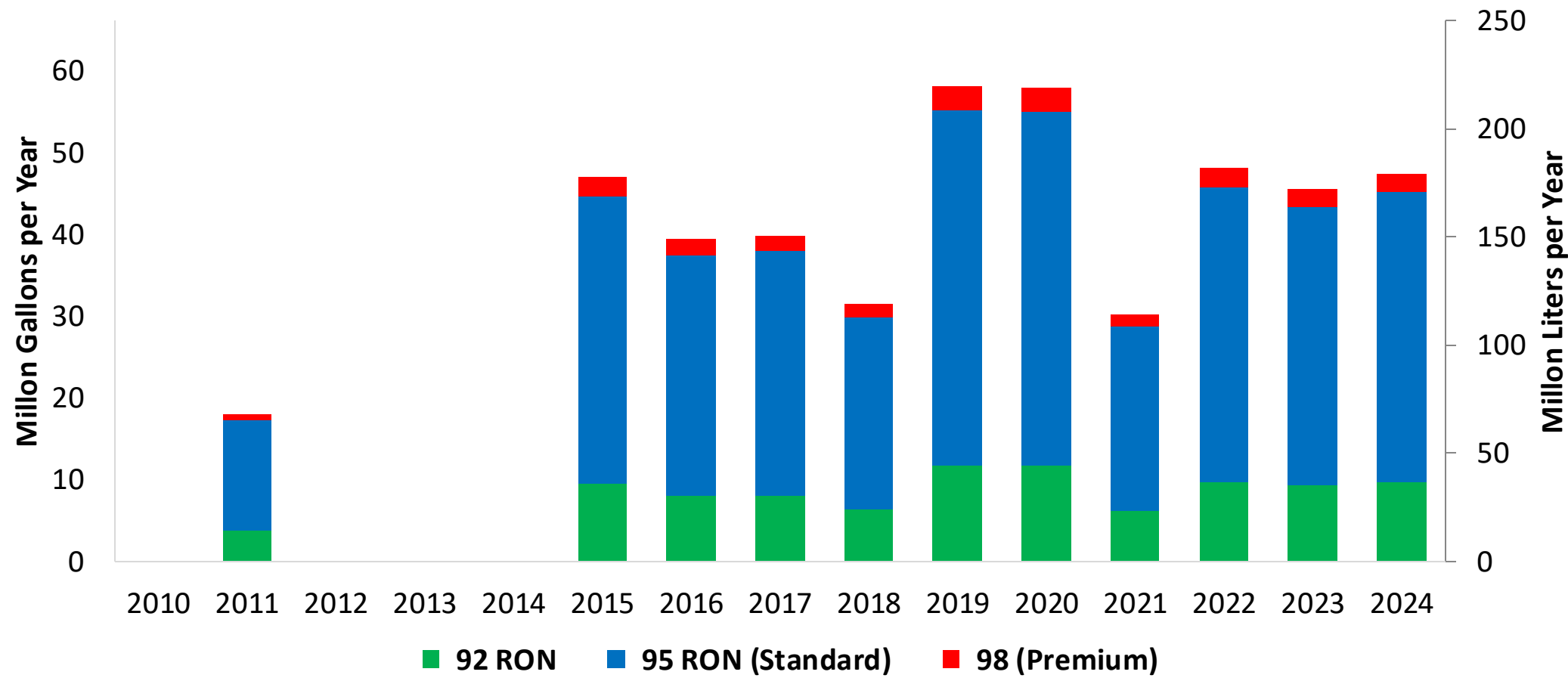
Gasoline Productionby Grade in Taiwan



Source: MOEAEA, E-STAT

Gasoline Imports in Taiwan

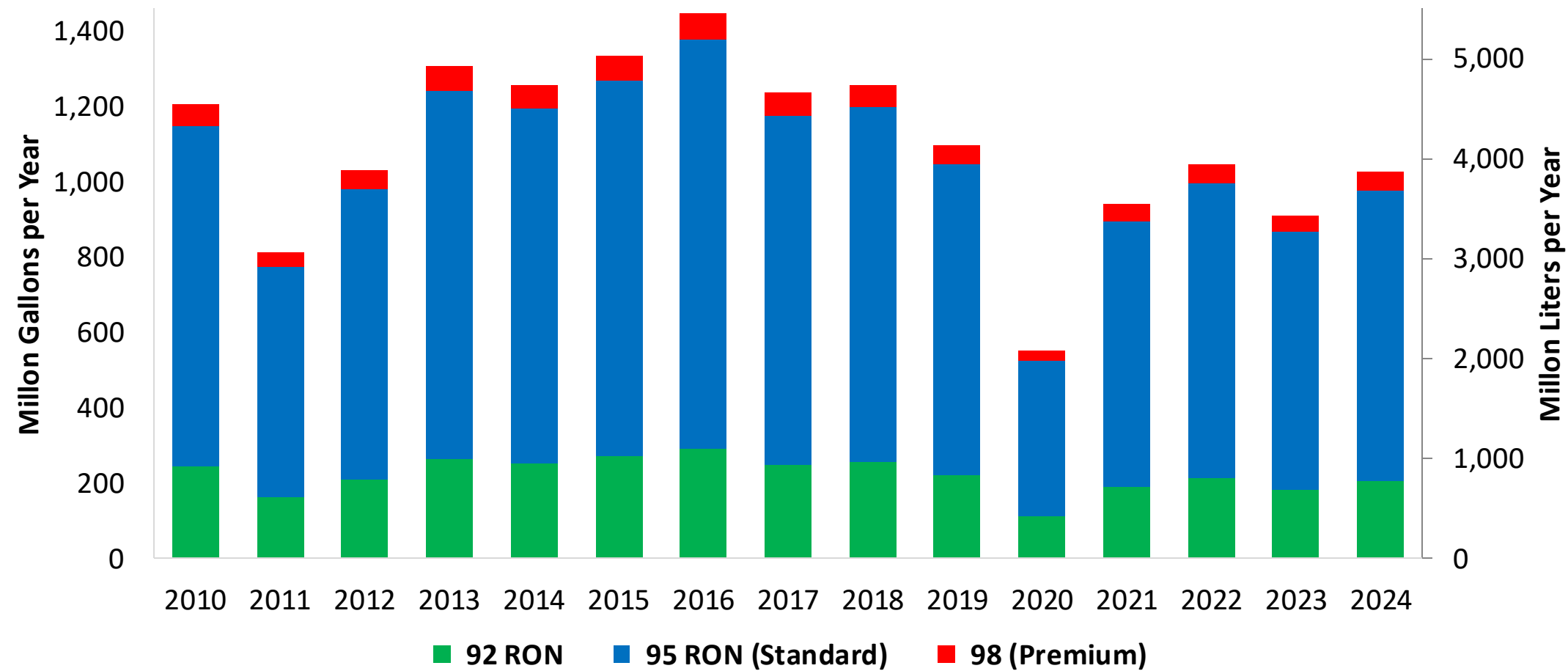
Gasoline Imports by Grade in Taiwan



Source: MOEAEA, E-STAT

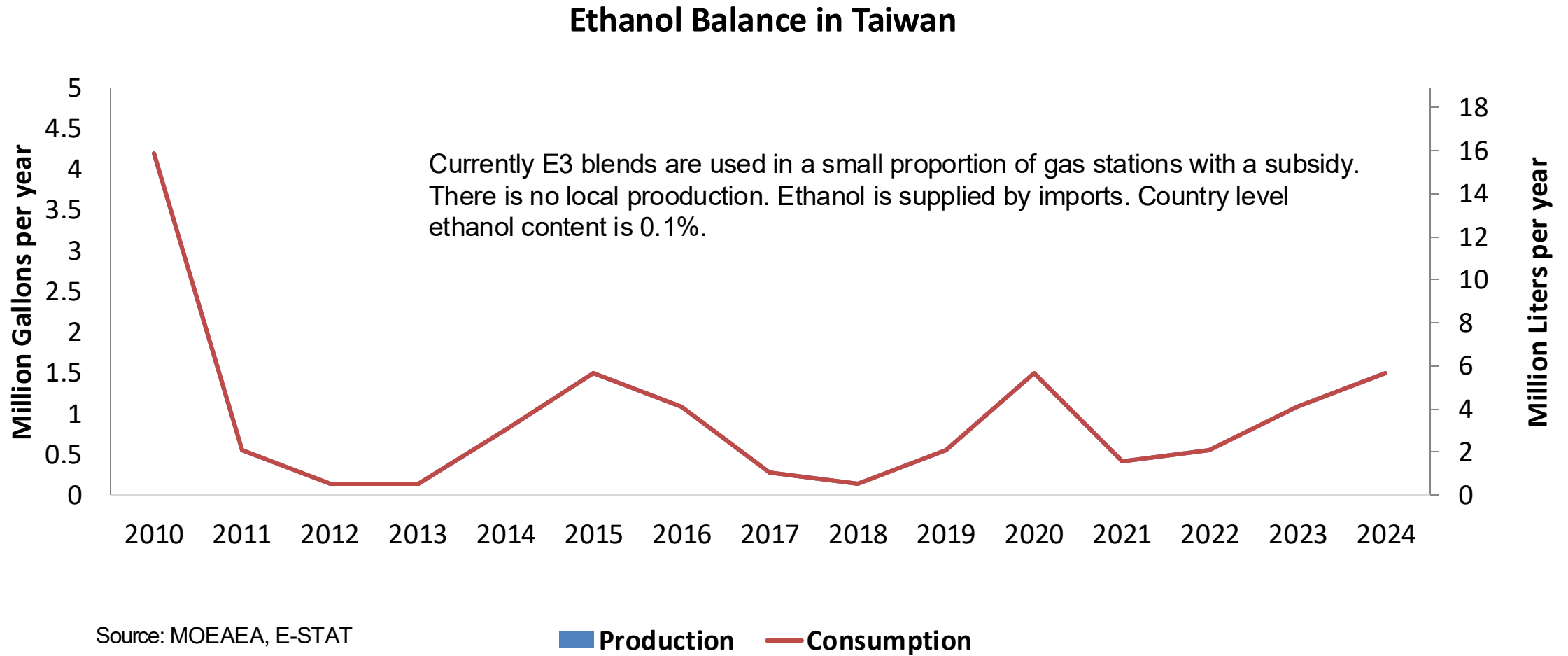
Gasoline Exports in Taiwan

Gasoline Exports by Grade in Taiwan



Source: MOEAEA, E-STAT

Ethanol Balance in Taiwan



Gasoline Quality in Taiwan

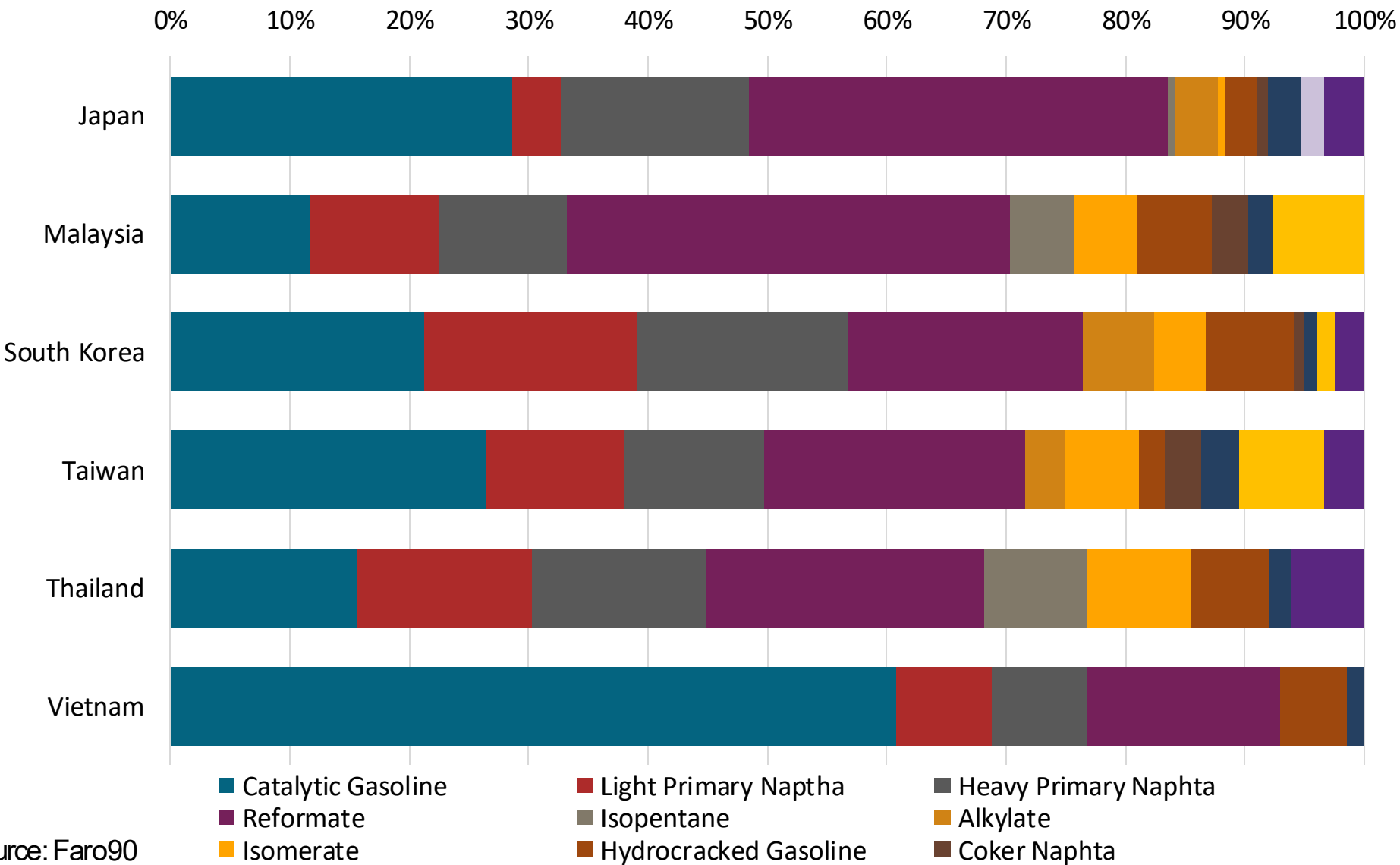
Name	Mobile Pollution Source Fuel Composition Control Standard Regulation					
Implementation Year	2024					
Applicability	Nationwide					
Grade	92 RON		95 RON (Standard)		98 RON (Premium)	
	Min	Max	Min	Max	Min	Max
RON	92.0		95.0		98.0	
MON						
AKI						
Benzene (%vol)		0.8		0.8		0.8
Aromatics (% vol)		35.0		35.0		35.0
Olefins (%vol)		18.0		18.0		18.0
Lead (g/L)						
Manganese (mg/L)						
Sulfur (ppm)		10		10		10
Oxygen (%v)		3.2		3.2		3.2
Ethanol (%vol)		10.0		10.0		10.0
MTBE (%vol)						
RVP (kPa)	45.0	66.9	45.0	66.9	45.0	66.9

Source: MOEAEA, MOENV, Formosa Petrochemical

Gasoline Component Blending in Asia

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Asia are:

- Catalytic Gasoline
- Light Primary Naptha
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics



Source: Faro90

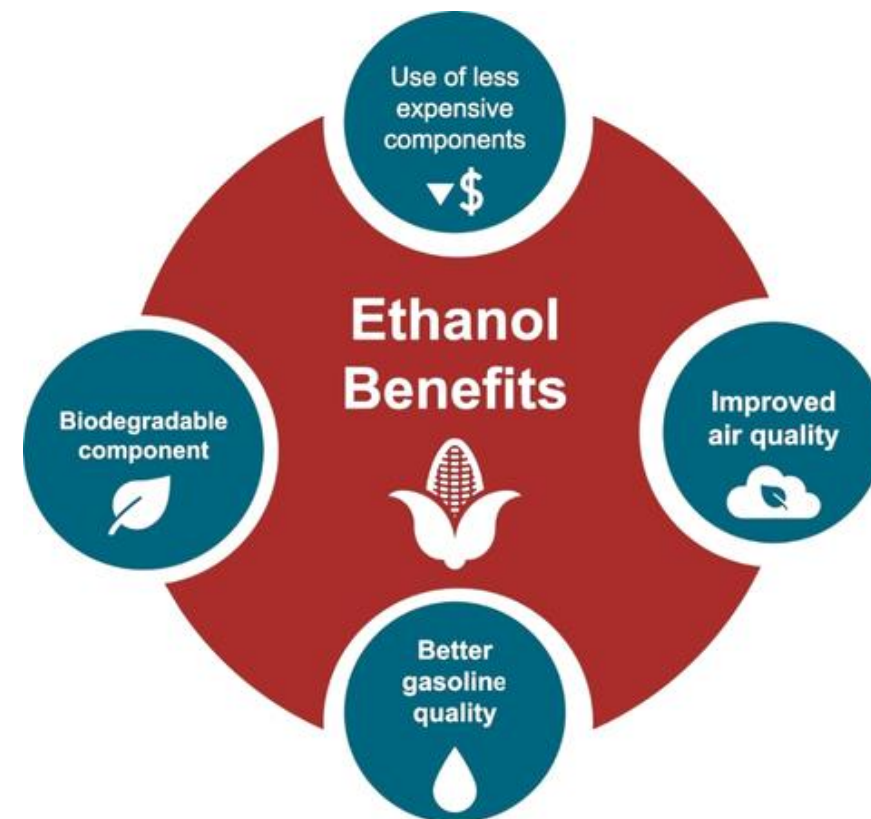
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

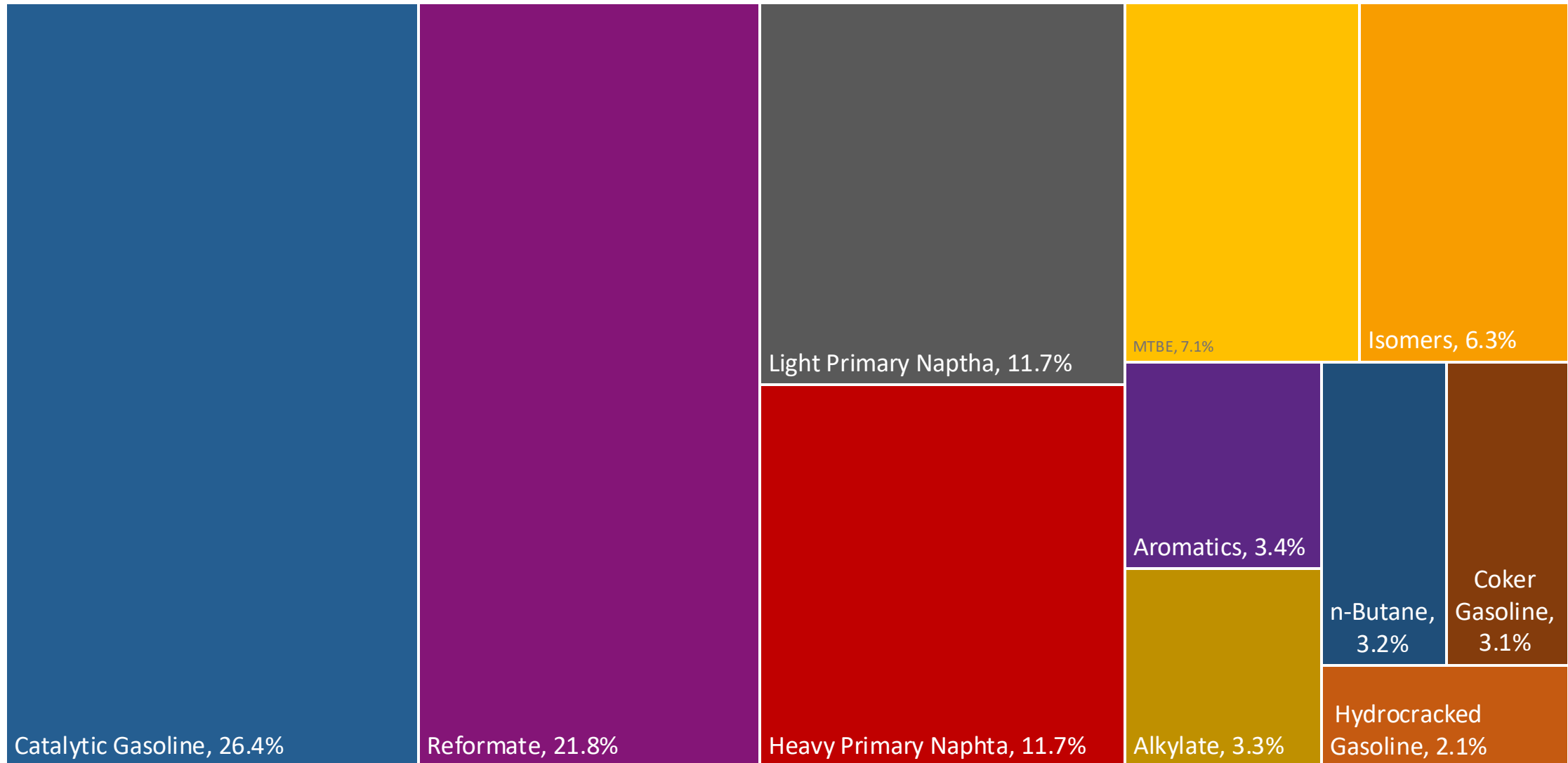
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

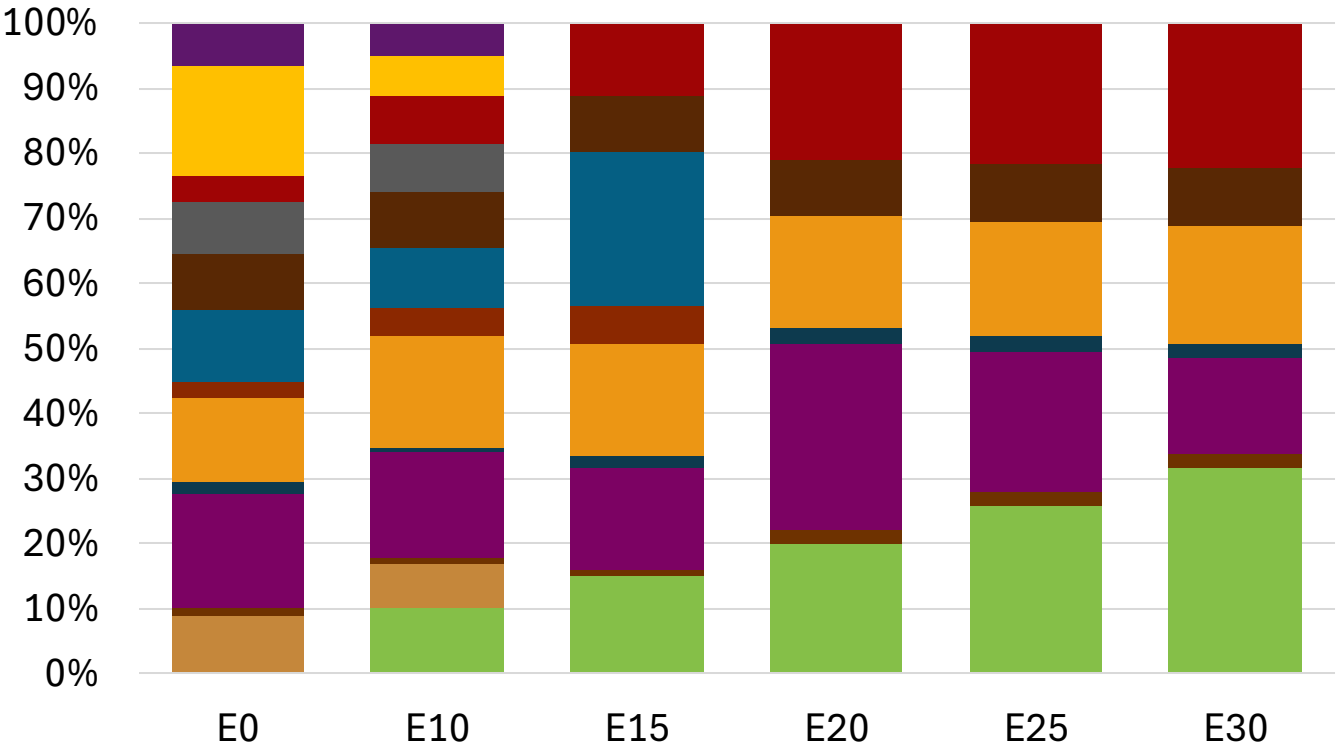
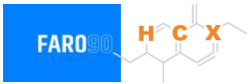
The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Taiwan Available Blending Components



Taiwan – Gasoline 95 – Constant Octane

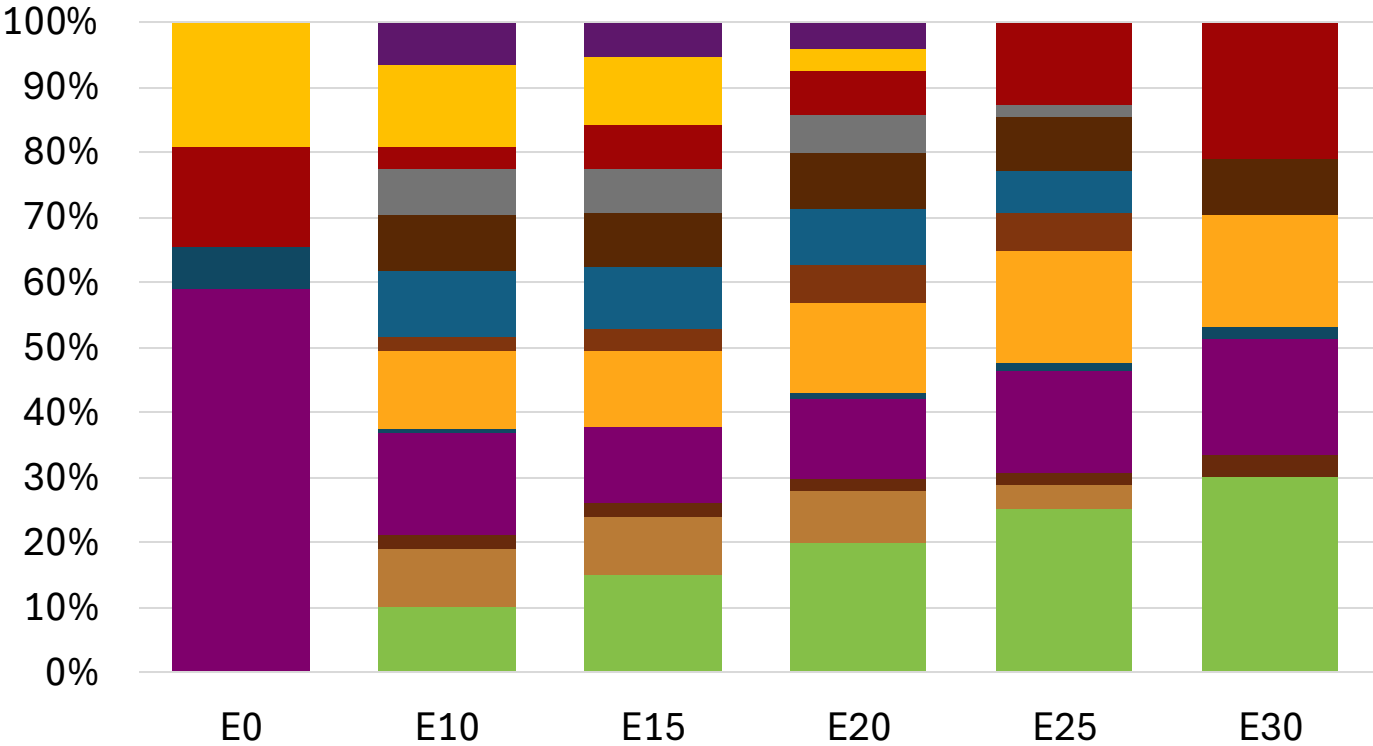
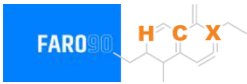


Average CIF 2024 Prices

Octane (RON)	95.0	95.0	95.0	95.8	95.0	95.0
Price (US\$/gal)	\$ 2.388	\$ 2.307	\$ 2.223	\$ 2.149	\$ 2.061	\$ 1.995

Source: Faro90

Taiwan - Gasoline 98 - Constant Octane

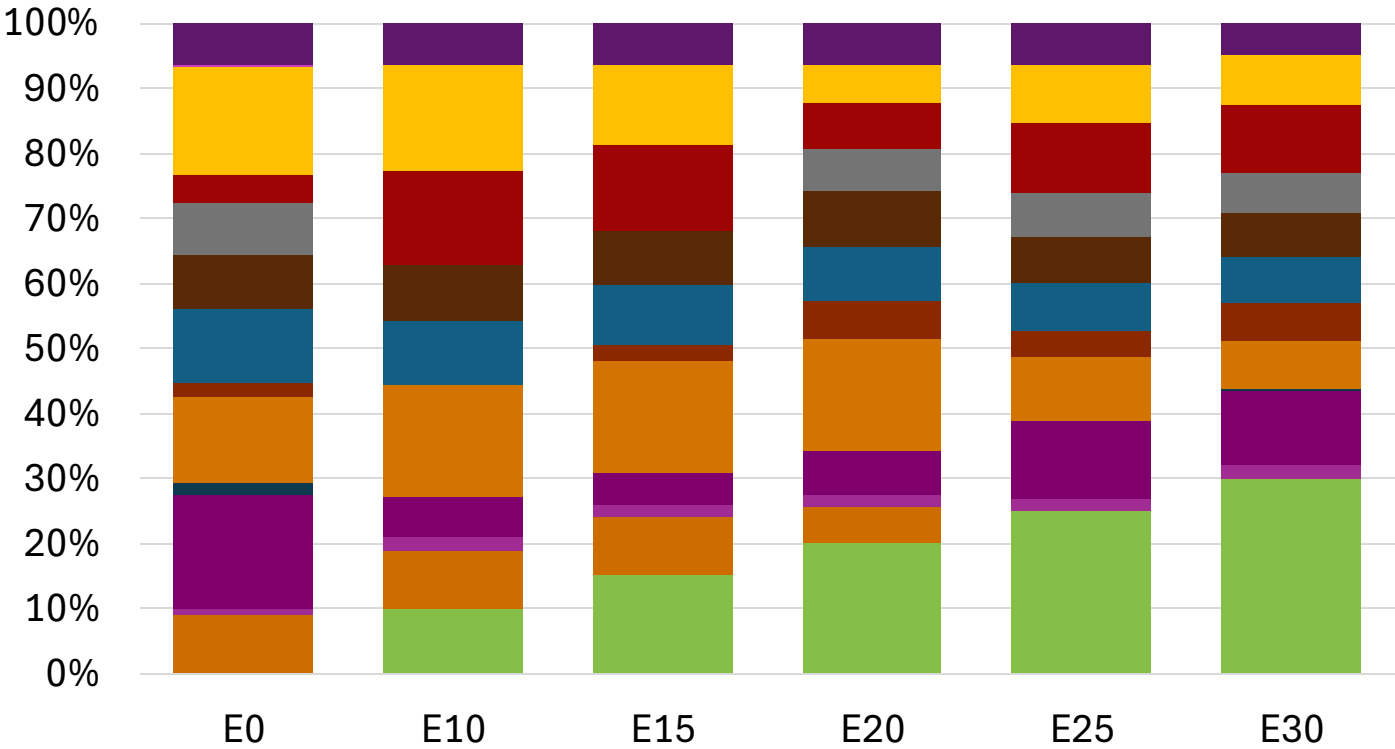
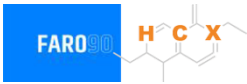


Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.6
Price (US\$/gal)	\$ 2.445	\$ 2.418	\$ 2.346	\$ 2.305	\$ 2.210	\$ 2.117

Source: Faro90

Taiwan – Gasoline 95 – Increased Octane

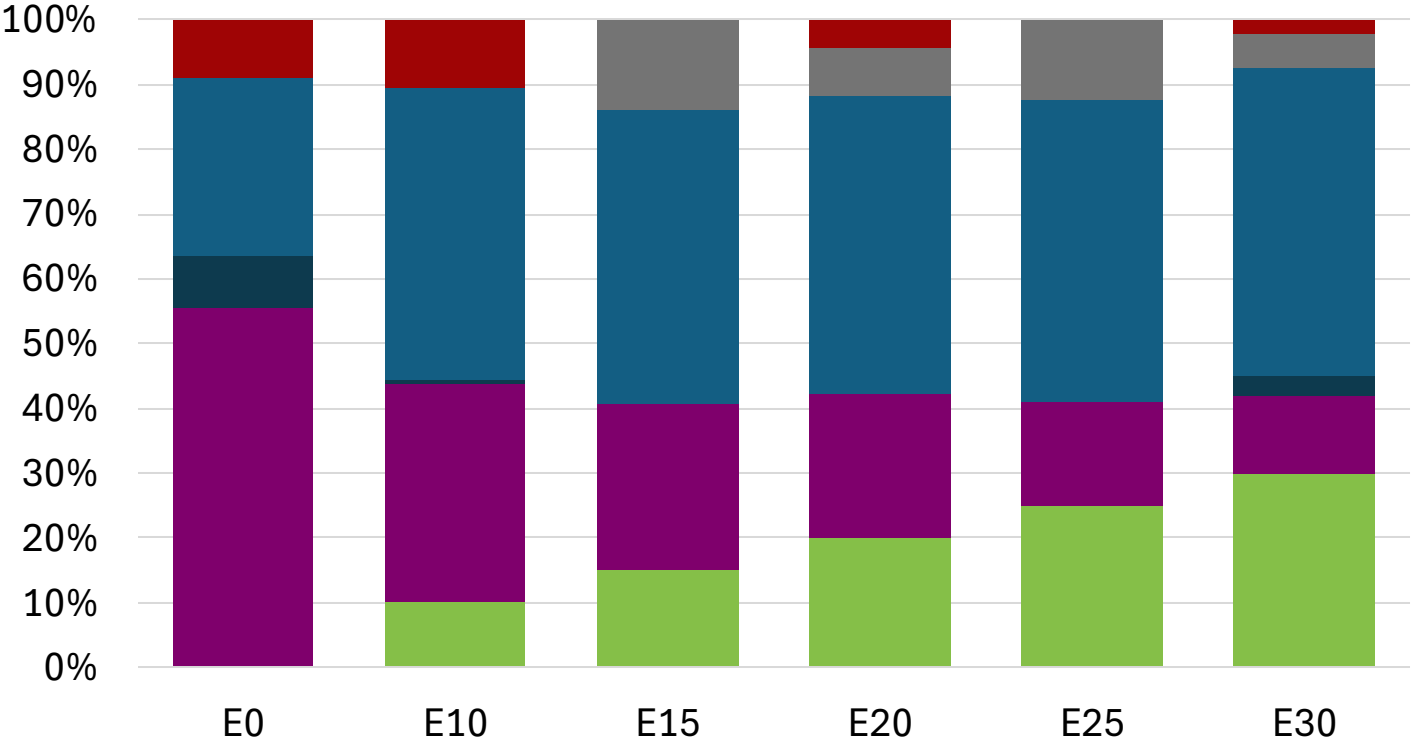
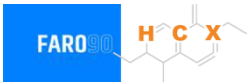


Average CIF 2024 Prices

Octane (RON)	95.0	96.4	97.4	98.1	100.6	102.0
Price (US\$/gal)	\$ 2.388	\$ 2.388	\$ 2.388	\$ 2.388	\$ 2.388	\$ 2.388

Source: Faro90

Taiwan – Gasoline 98 – Increased Octane



Average CIF 2024 Prices

Octane (RON)	98.0	99.8	99.9	101.3	102.6	103.9
Price (US\$/gal)	\$ 2.445	\$ 2.445	\$ 2.445	\$ 2.445	\$ 2.445	\$ 2.445

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

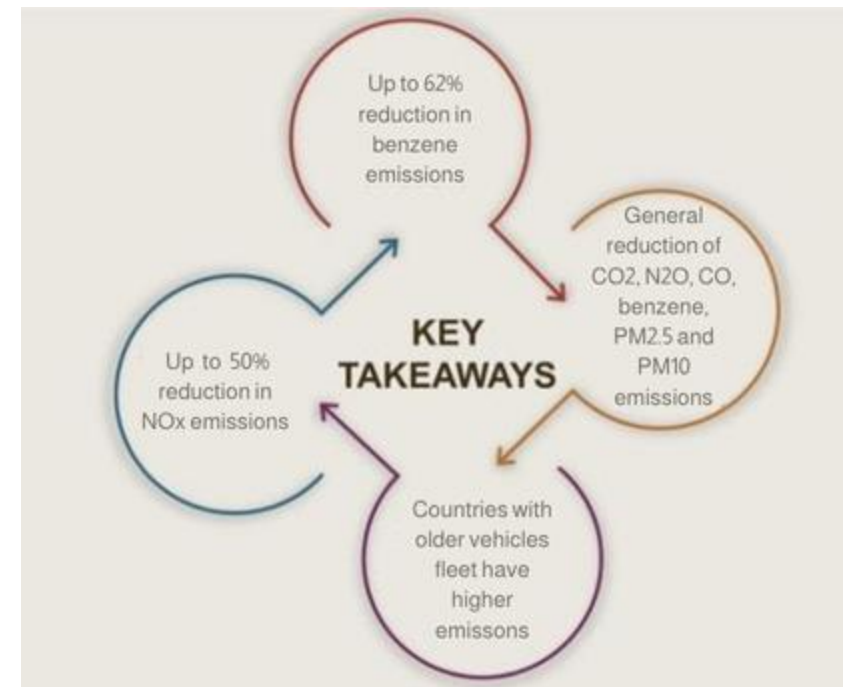
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

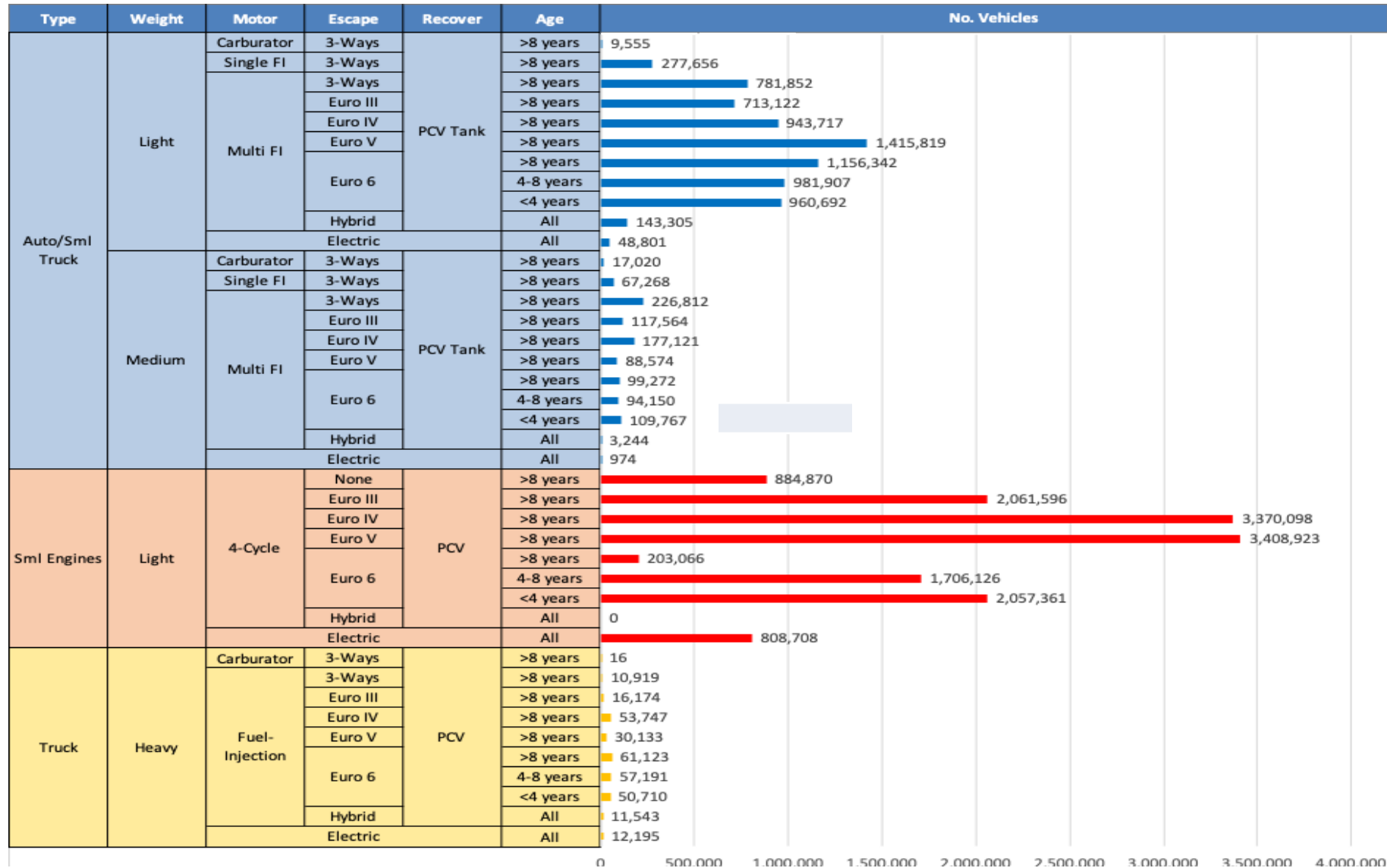
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

*Source: Bureau of transportation statistics.

Main Results



Gasoline Vehicle Fleet - Taiwan

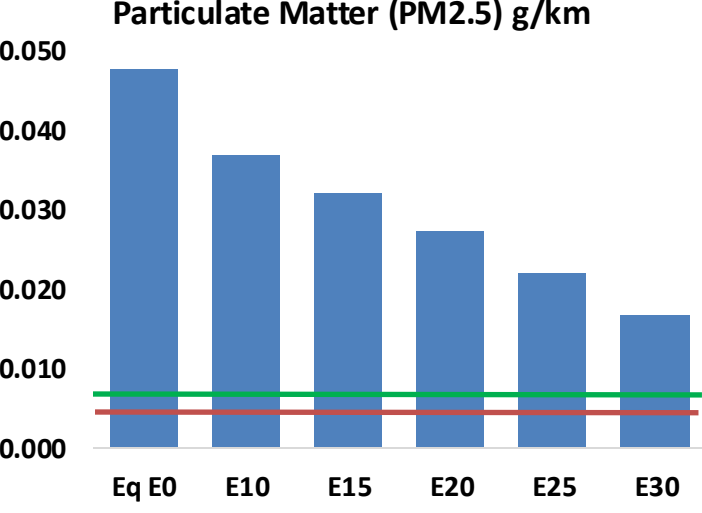
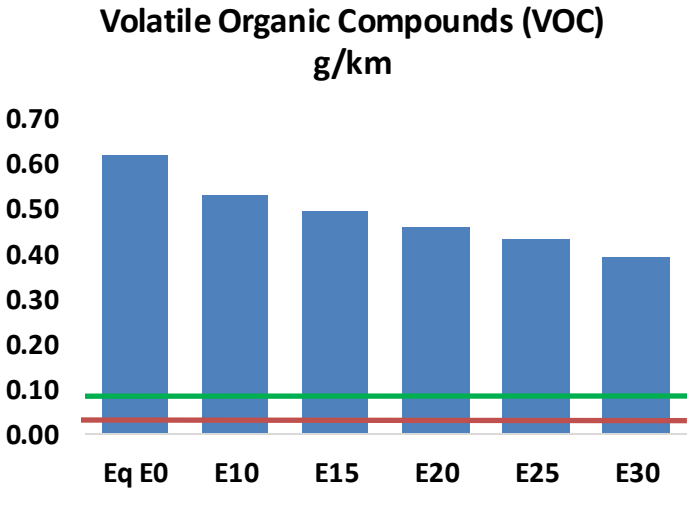
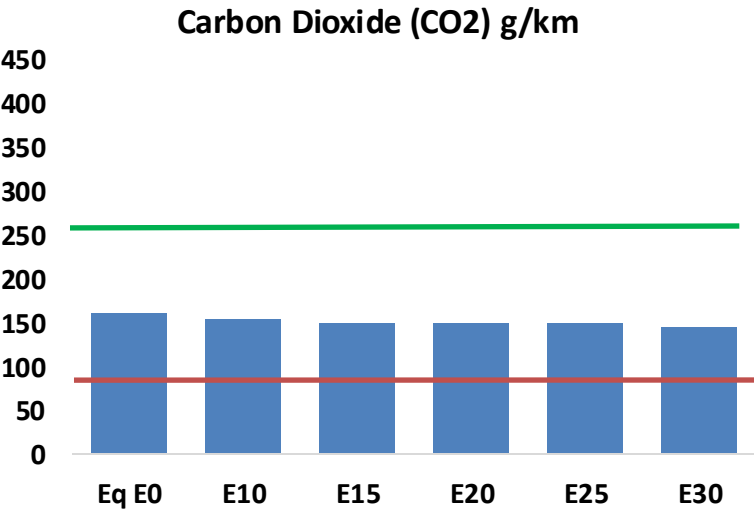
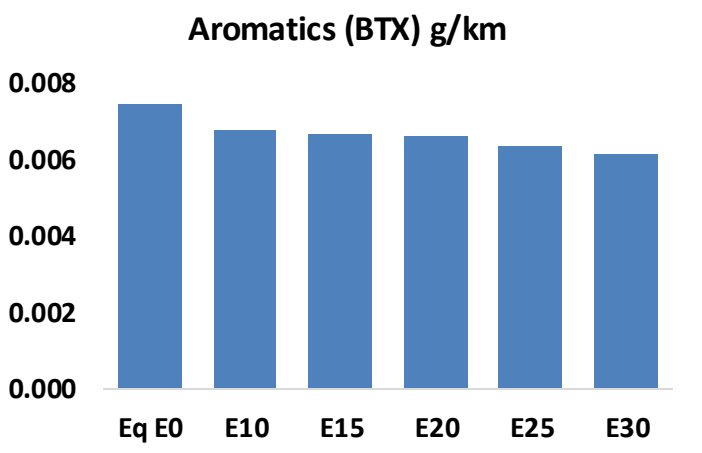
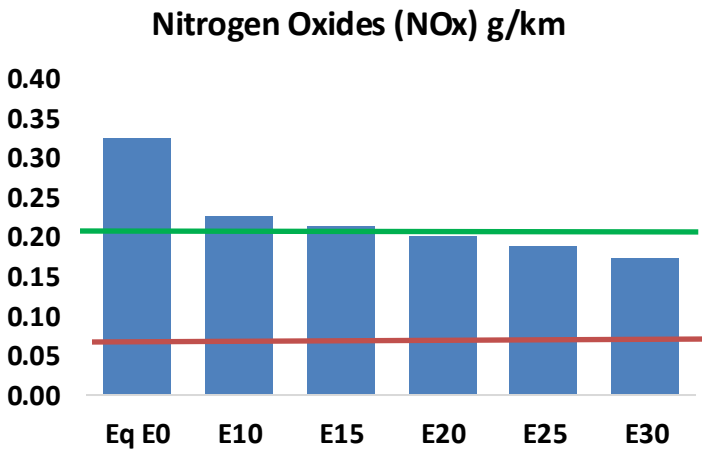
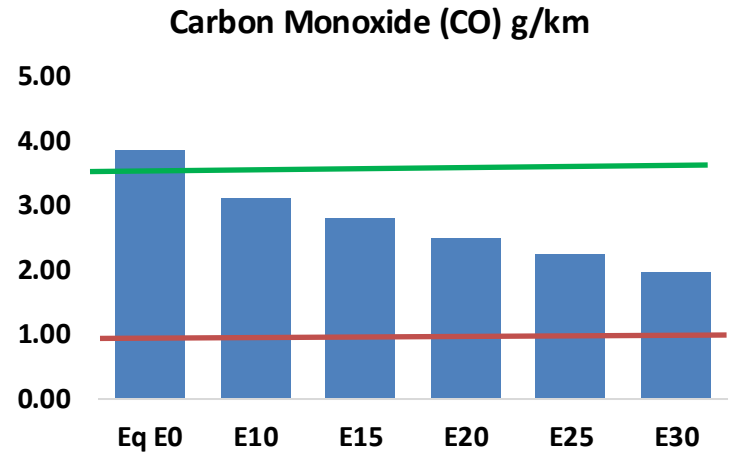
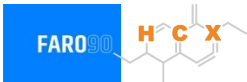


Vehicle Fleet: 23,239,031
Gasoline Vehicle Fleet: 20,069,210

Source: DATA-GOV.tw, CIECData

Taiwan – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



Taiwan – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,360,185	1,230,696	1,101,207	990,234	876,319	788,050	691,909	-19%	-36%
CO2	57,083,319	55,615,462	54,229,153	53,138,861	52,599,342	52,408,803	51,442,758	-5%	-8%
VOC	218,167	202,516	186,866	174,260	161,676	152,029	139,049	-14%	-26%
NOx	114,877	86,158	80,414	75,790	71,477	66,721	61,689	-30%	-38%
SOx	636	588	540	499	460	418	373	-15%	-28%
NH3	23,536	23,303	23,070	23,179	23,440	23,623	23,776	-2%	0%
N2O	766	762	759	763	779	787	796	-1%	2%
CH4	54,499	54,499	54,499	55,317	56,516	57,551	58,532	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	1,595	1,496	1,397	1,321	1,247	1,190	1,112	-12%	-22%
Acetaldehyde	4,628	5,312	7,794	11,869	16,170	18,477	21,833	68%	249%
Formaldehyde	18,061	17,559	20,469	24,035	25,180	27,856	30,325	13%	39%
Benzene	2,632	2,529	2,397	2,359	2,339	2,237	2,165	-9%	-11%
PM2.5	3,932	3,491	3,051	2,649	2,250	1,826	1,383	-22%	-43%
PM10	12,864	11,422	9,980	8,667	7,360	5,972	4,526	-22%	-43%

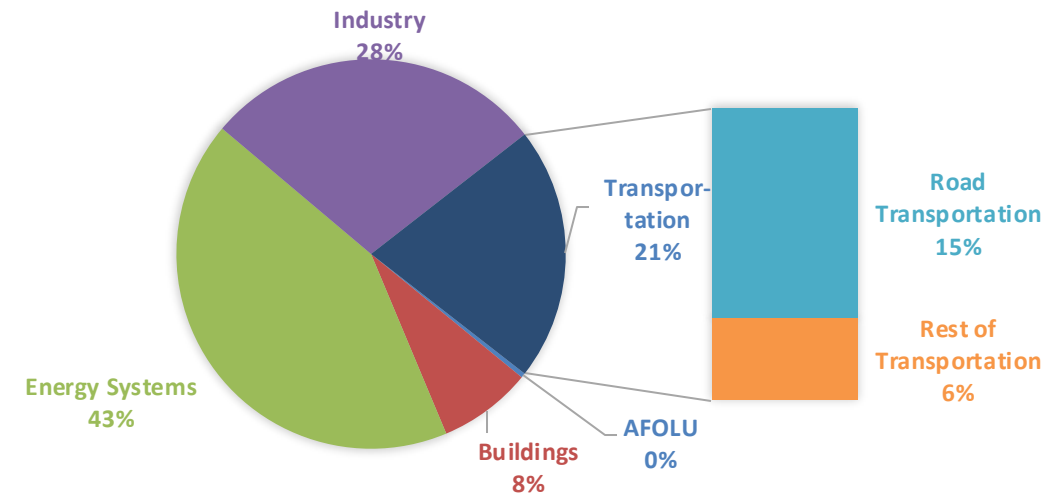
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

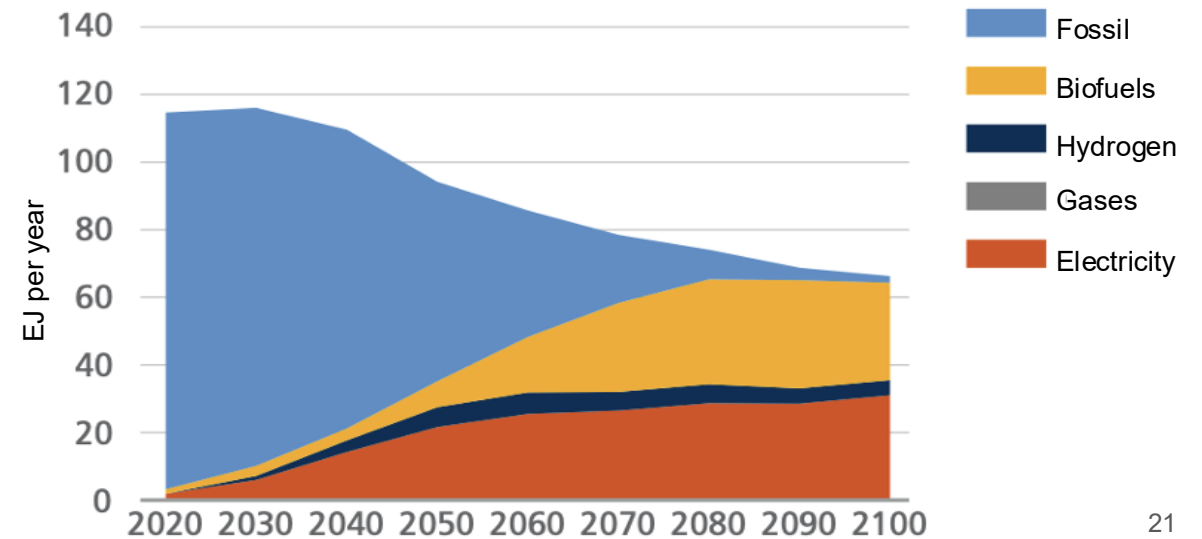
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

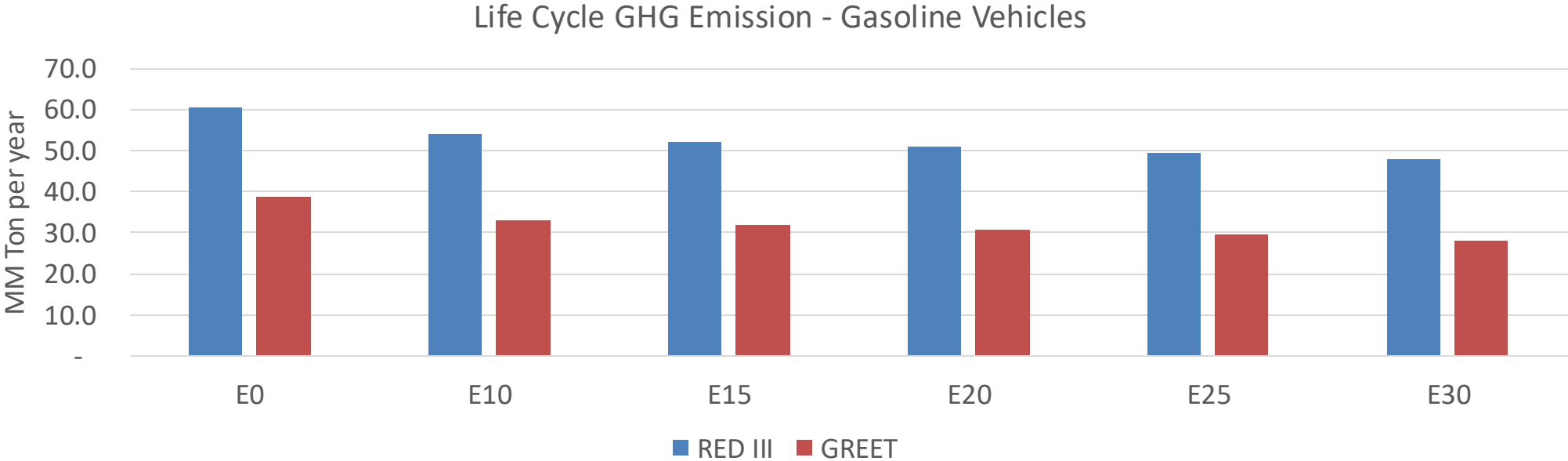
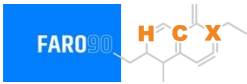
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Taiwan – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	291.7
Target @ 2035 (MMT/year)	204.2
Delta	87.5

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	14.6%	18.0%	20.9%	23.6%	27.2%
RED II/III	0.0%	10.6%	13.4%	15.8%	18.0%	20.8%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	6.4%	8.0%	9.2%	10.4%	12.0%
RED II/III	0.0%	7.3%	9.2%	10.9%	12.4%	14.4%

Source: Greet, RED II/III, climateactiontracker.org, F90